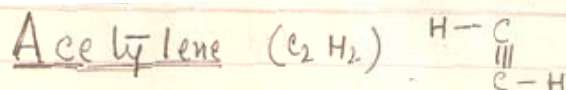




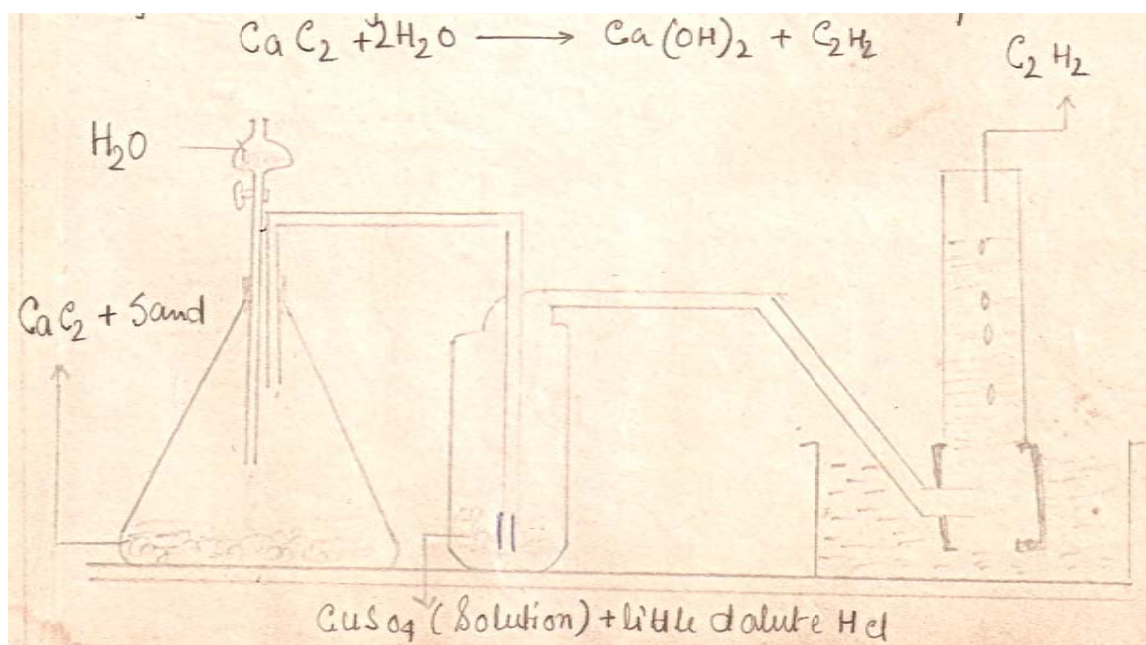
General Formula
 $C_n H_{2n-2}$



Preparation $\rightarrow$ Acetylene is prepared in the laboratory by treating Calcium Carbide with water/dilute HCl at ordinary temperature



Procedure $\rightarrow$ In a conical flask about 50 gms of CaC_2 (Calcium Carbide) is taken, it is mixed with a little sand to slow down the chemical reaction. The flask is fitted with a dropping funnel through which water is poured drop by drop. It is fitted with a delivery tube through which C_2H_2 gas passes into a washing bottle containing $CuSO_4$ solution and a little dilute HCl. The other end of the washing bottle is connected with another delivery tube as shown in the figure to pass the gas to the gas jar.





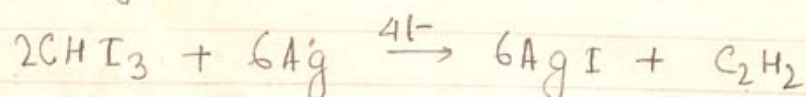
On adding water C_2H_2 is evolved which is purified and collected by the downward displacement of water.

Precaution $\rightarrow$ As the reaction is very rapid a little sand is added to Calcium Carbide (CaC_2) so that the reaction is slow.

Purification $\rightarrow$ The impurities are NH_3 , PH_3 (Phosphine) H_2S , AsH_3 (Arsenic hydride / Arsine gas) because CaC_2 is an industrial product so the gas is passed through acidified $CuSO_4$ solution.

Other Methods Of Preparation

When Iodoform (Haloform) is heated with Silver powder we get Acetylene. (R.U)



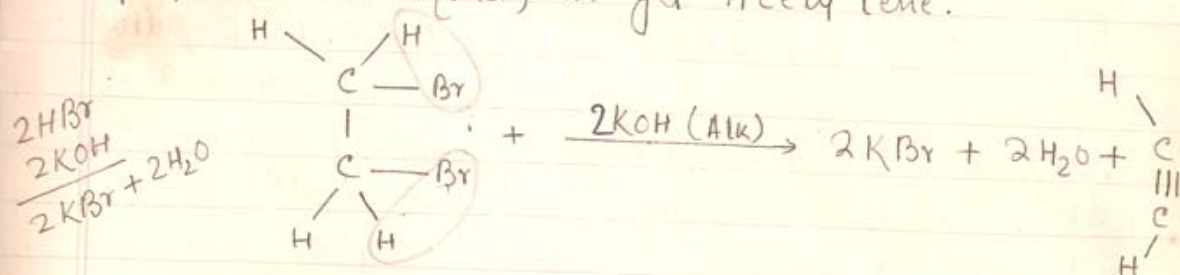
Synthesis of Acetylene

When Carbon is sparked in Hydrogen gas Acetylene is formed. In a hard glass globe two graphite electrodes high volt- electricity is passed in H_2 gas, atmosphere C_2H_2 is formed as shown in the figure.



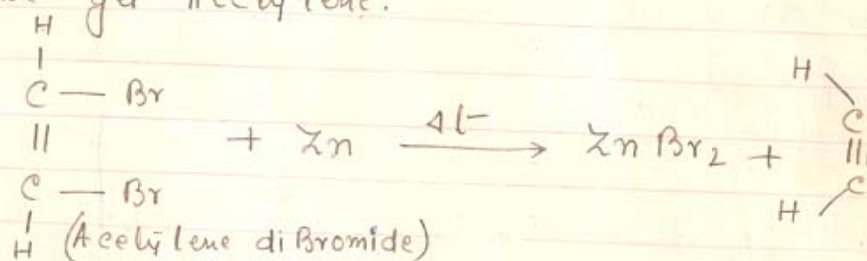
3. Conv

When Ethylene Bromide is treated with KOH solution prepared with alcohol, we get Acetylene.



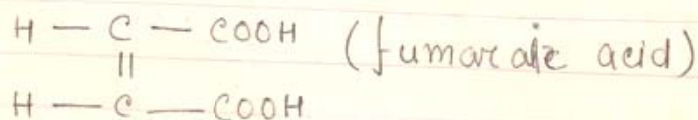
W.H

④ When Acetylene di Bromide is heated with zinc dust we get Acetylene.

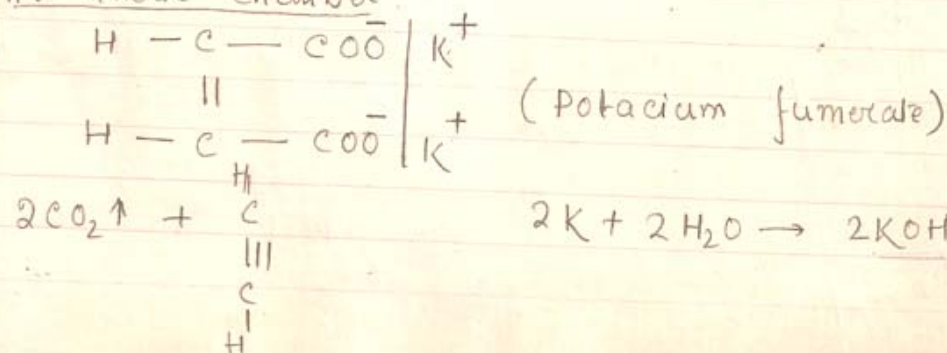


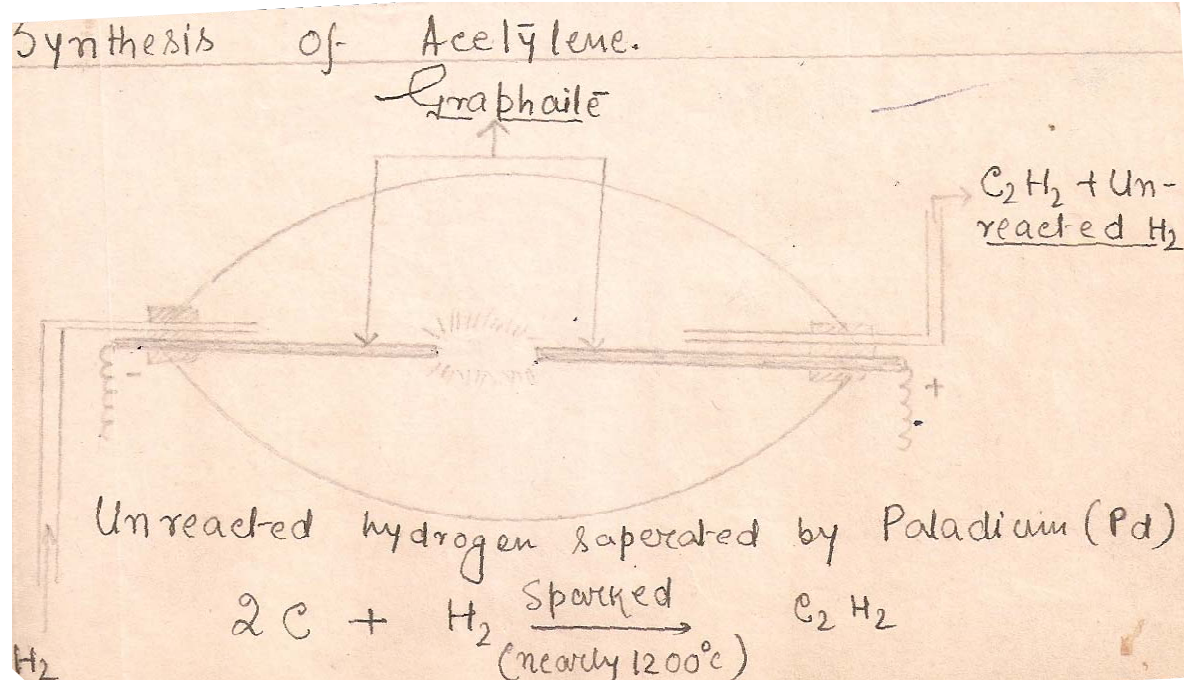
Compt

⑤ When Potassium fumerate solution is electrolysed (Kolbe's Synthesis) we get Acetylene and Carbon dioxide at Anode. CO_2 is removed by passing through KOH solution to get pure C_2H_2 (fig. as dis. curve).



At Anode Chamber





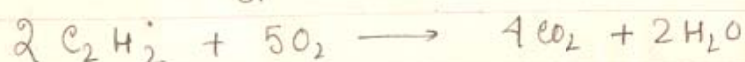
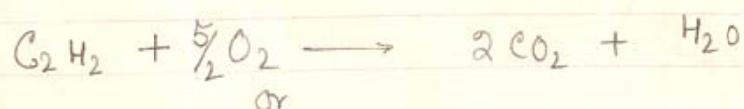


Physical properties of Acetylene.

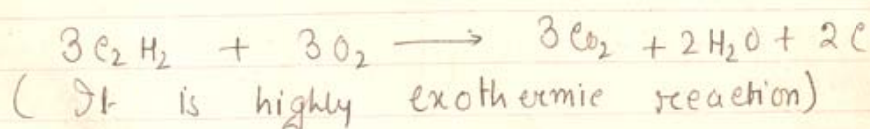
It is a colourless gas and has a peculiar smell similar to garlic. It is lighter than air, slightly soluble in water but highly soluble in Alcohol and Acetone, It is converted into a liquid at low temperature. It reacts catches fire if ignited. It is stored in cylinder under high pressure in which small bricks soaked with Acetone are kept.

Chemical properties

1. Combustion/oxidation $\rightarrow$ It burns with a smoky flame but if mixed with oxygen and burnt it gives a greenish blue of light.

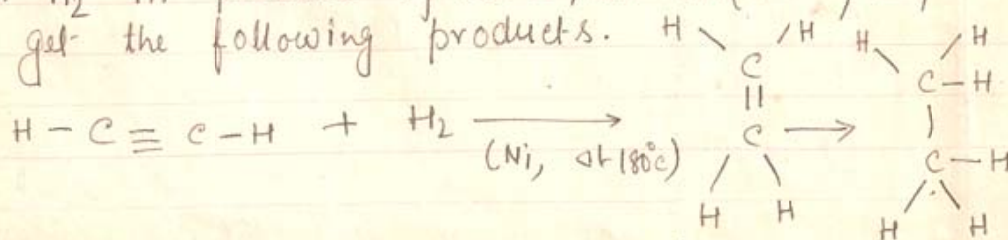


If the oxygen is less we get a little Carbon



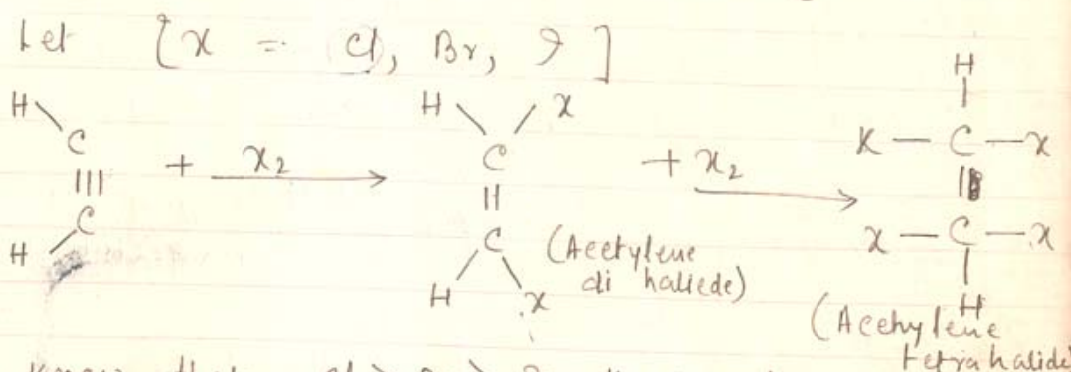
Conv

2. With Hydrogen $\rightarrow$ When Acetylene gas is heated with H_2 in presence of catalyst like Ni/Pt/Pd we get the following products.





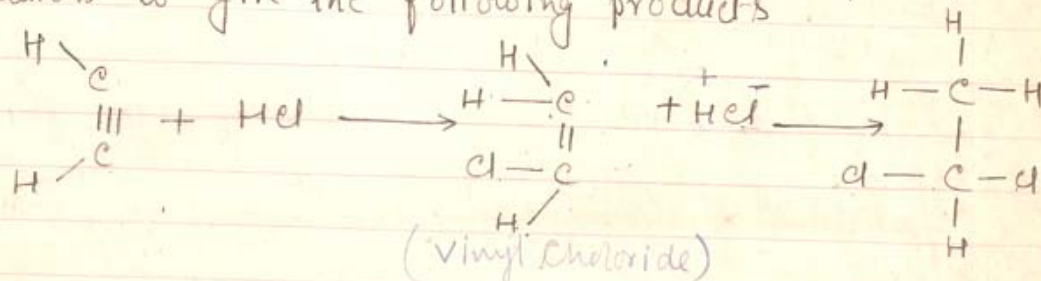
3. With Hologens $\rightarrow$ Cl, Br, I react with Acetylene to form Acetylene tetra halides as given below



We know that $\text{Cl} > \text{Br} > \text{I}$ therefore to get Acetylene tetra chloride we are to pass Acetylene gas and chlorine gas through water because Cl_2 and C_2H_2 formed Carbon particle at room temperature

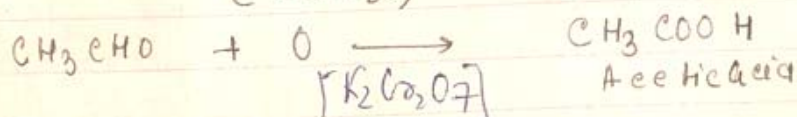
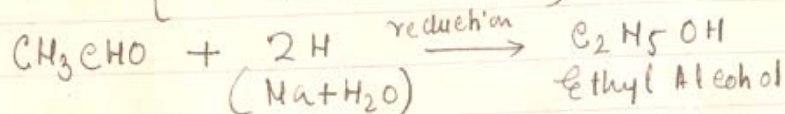
$$\text{C}_2\text{H}_2 + \text{Cl}_2 \longrightarrow 2\text{HCl} + 2\text{C}$$

- ④ With HCl $\rightarrow$ Acetylene reacts with HCl/HBr/HI solutions to give the following products



"Markownikoff's rule"

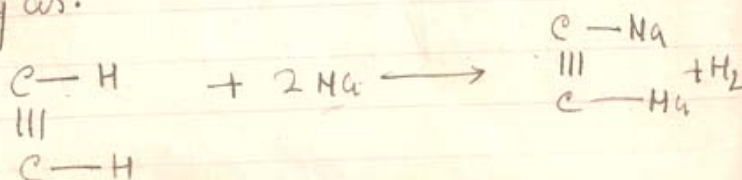
In case of a double bond existing between two Carbon atoms which are linked to unequal number of hydrogen atom, the addition of molecule, takes place in such a way that the negative part of the adding molecule goes to the Carbon atom which contains small number of hydrogen atom. Example is given below

$$\text{C}_2\text{H}_2 \xrightarrow[\text{at } 80^\circ\text{C}]{\text{20\% H}_2\text{SO}_4, \text{1\% HgSO}_4, \text{Under pressure}} + \text{H}_2\text{O} \quad \text{CH}_3\text{CHO} \text{ Acetaldehyde}$$


C1=CC=CC=C1

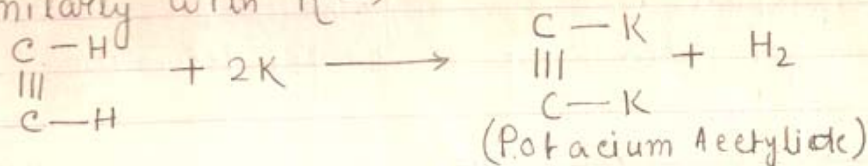


7. With Sodium $\rightarrow$ When Acetylene gas is passed through heated Sodium (Na) or K we get hydrogen (H_2) gas.

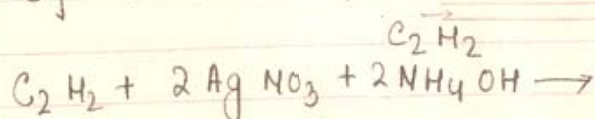




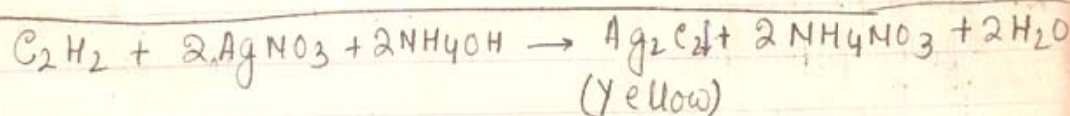
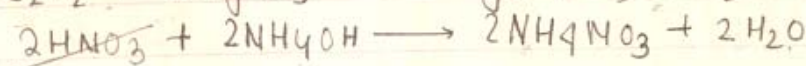
Similarly with K →



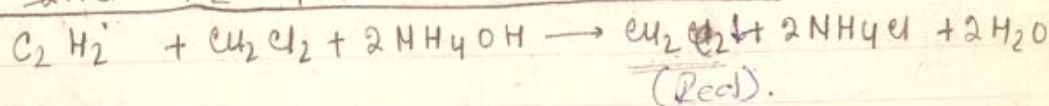
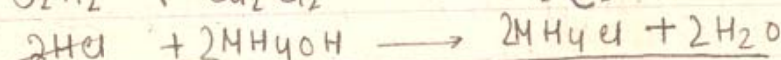
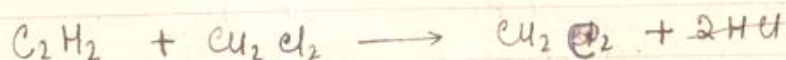
- ⑧ ^{W.H} With AgNO_3 Solution alkaliified with NH_4OH → When Acetylene gas is bubbled through Ammoniacal AgNO_3 Solution we get yellow precipitation of Silver Acetylide.



$\text{AgNO}_3 + \text{NH}_4\text{OH}$
(Ammoniacal AgNO_3)



- ⑨ With Ammoniacal cuprous chloride → When Acetylene bubbled through Ammoniacal cuprous chloride Solution we get red precipitation of copper Acetylide.



Uses of Acetylene

- ① To ripen fruits before due time.
2. Oxy-Acetylene gas flame is used for cutting, welding and boring of metals.



3. To prepare Benzene Synthetically.
4. To get dazzling light for decoration.

Qv. You are given three gas jars containing C_2H_6/CH_4 , C_2H_4 , C_2H_2 how would you identify each of the gas jar.

Reagent or Experiment-	C_2H_2	C_2H_4	C_2H_6/CH_4
1. Three pieces of blotting paper dipped in Amonical $AgNO_3$ Solution and introduced in each gas jar.	turns yellow	no change	no change
2. Two pieces of filter paper dipped in Br_2 water, are introduced in gas jars		The brown colour become colourless	brown colour remains as it is.

All the three gases are colourless therefore we can not distinguish by colour.

Three pieces of blotting paper dipped in Amonical $AgNO_3$ Solution and then introduced in each gas jar. In which the blotting paper turns yellow, this gas must be Acetylene. This gas is removed.

Now two pieces of blotting paper dipped in Bromine water (brown) are introduced in two gas jars. In which the brown colour is discharged, this gas must be (C_2H_4) Ethylene. The left-gas jar contains C_2H_6/CH_4 .



Qv. Distinguish among Methane ($\text{CH}_4/\text{C}_2\text{H}_6$), Ethylene (C_2H_4), and (C_2H_2) Acetylene in a tabular form.

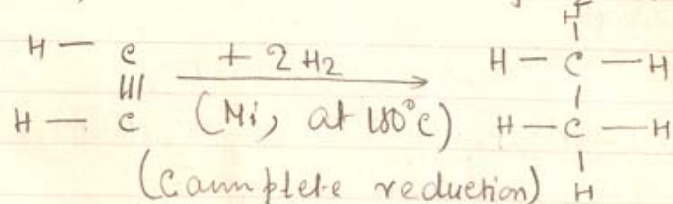
Reagent	$\text{CH}_4/\text{C}_2\text{H}_6$	C_2H_4	C_2H_2
1. Ammonical AgNO_3 Solution.	No change	No change	Turns yellow
2. Ammonical Cu_2Cl_2	No change	No change	Turns red.
3. Br_2 water	No change	Colourless	Colourless
4. With Con. H_2SO_4	No reaction	Ethyl hydrogen Sulphate is formed.	Acetaldehyde is formed
5. With KMnO_4 Solution	No change	Pink colour of KMnO_4 goes Glycol is formed	Pink colour goes.

Inter Conversion

Acetylene to Ethane and vice versa:-

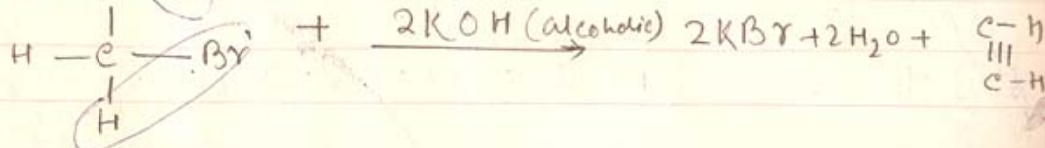
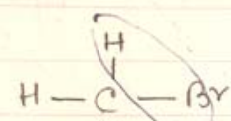
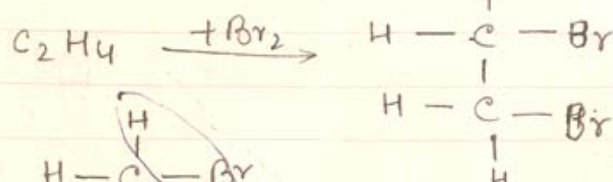
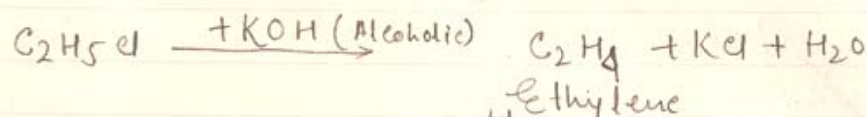
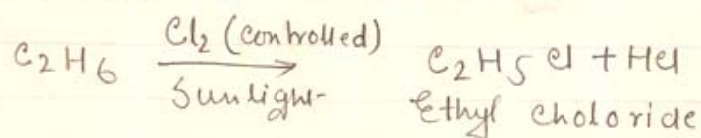
Ans

① When Acetylene gas is heated with H_2 in the presence of Ni as a catalyst at about 180°C , due to addition of H_2 we get Ethane.



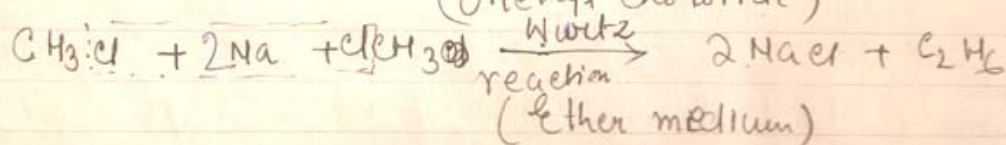
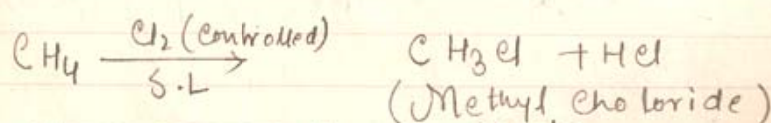
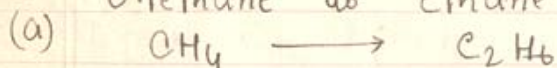


(b) Ethane to Acetylene



When Ethylene gas is treated with Cl_2 gas in a controlled way, in the presence of sunlight, Ethyl chloride is formed. This ethyl chloride is added with alcoholic KOH we get Ethylene. This is treated with Br_2 to get ~~Ethylene~~ ^{Ethylene} bromide. When Ethylene Bromide is treated with alcoholic KOH , Acetylene is obtained.

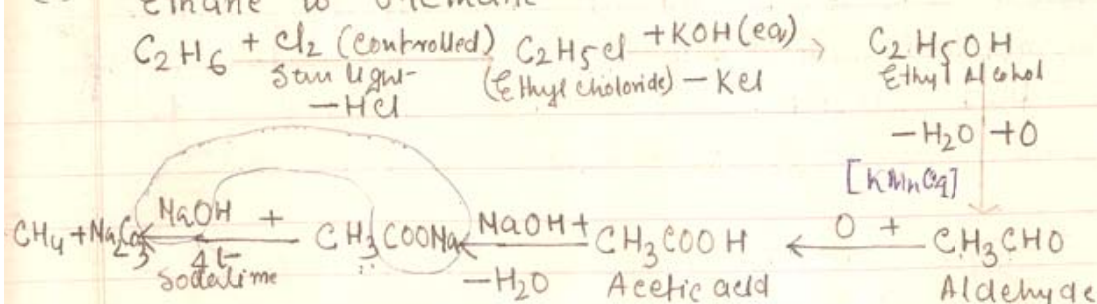
2. Methane to Ethane and vice versa





When Methane gas and Cl_2 gas mixture is kept in sunlight (controlled) we get Methyl chloride. When Methyl chloride and metallic Na is heated in ether medium we get Ethane.

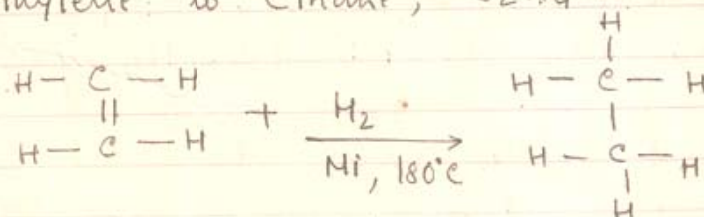
(b) Ethane to Methane $\rightarrow$



When Ethane is mixed with Cl_2 in a controlled way we get Ethyl chloride. This is treated with KOH(aq) solution to get Ethyl Alcohol which on oxidation gives Acetaldehyde and on further oxidation, Acetic acid is formed. When this is treated with NaOH solution and evaporated we get solid Sodium Acetate. When a mixture of Sodium Acetate and Sodalime is heated we get Methane.

3. Convert Ethylene to Ethane and vice versa

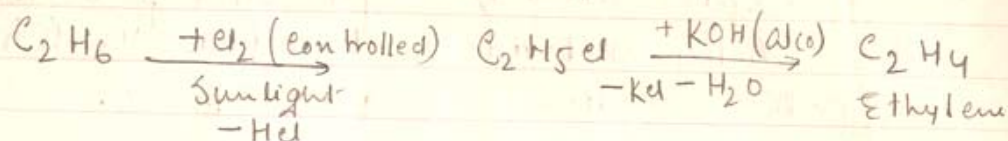
(i) Ethylene to Ethane, $\text{C}_2\text{H}_4 \rightarrow \text{C}_2\text{H}_6$



When Ethylene is heated with hydrogen in presence of Ni, as a catalyst due to addition reaction of H_2 we get Ethane.

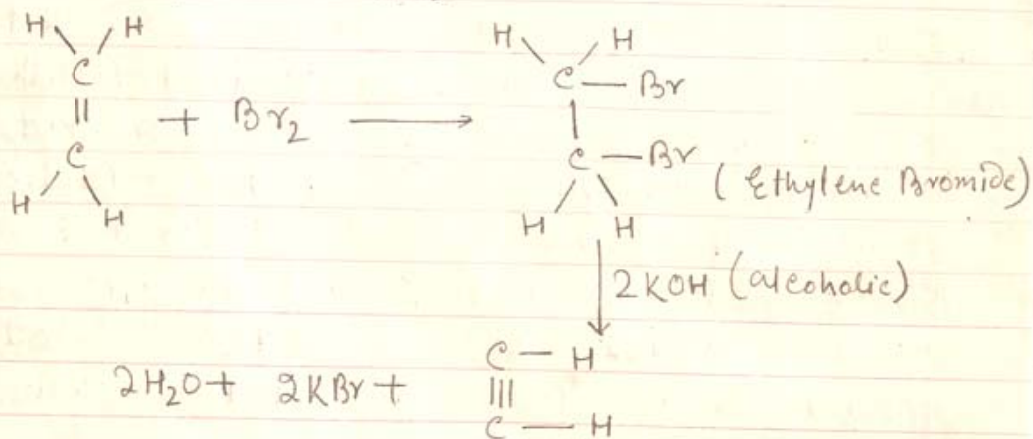


(b) Ethane to Ethylene

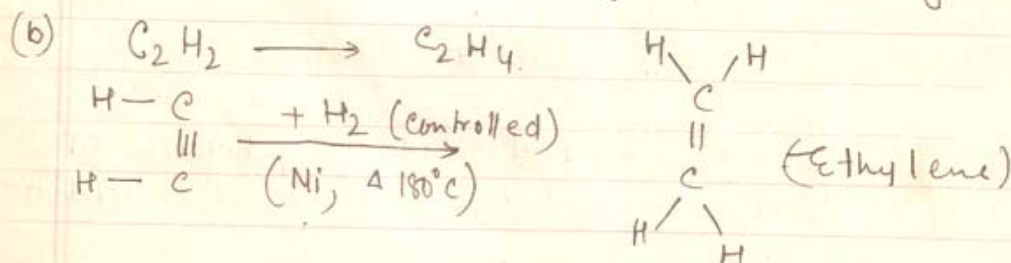


When Ethane is mixed with Cl_2 gas in a controlled way in the presence of S.L we get Ethyl Chloride. When Ethyl Chloride is treated with alcoholic KOH , Ethylene is formed.

4.(a) Ethylene to Acetylene and vice versa $\rightarrow$
 $\text{C}_2\text{H}_4 \rightarrow \text{C}_2\text{H}_2$



When Ethylene is mixed with Br_2 we get Ethylene Bromide. On treating Ethylene Bromide with alcoholic KOH we get Acetylene gas.

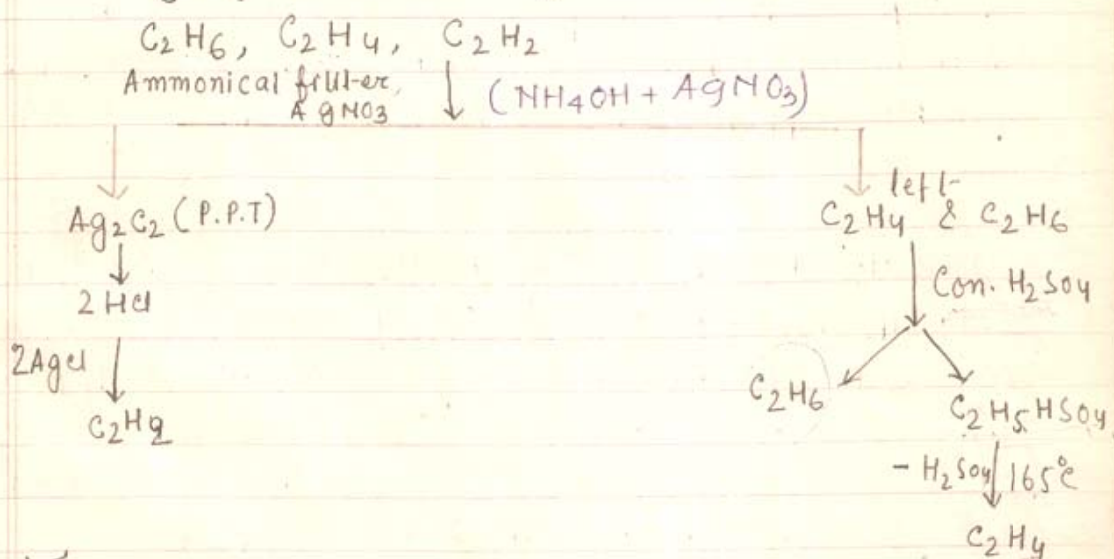




When Acetylene is heated with hydrogen in a very controlled way in the presence of Ni, at 180°C due to addition of hydrogen we get Ethylene.

COMPT

You are given a mixture of C_2H_6 , C_2H_4 , C_2H_2 in a gas-jar how can you separate each of them.



The mixture of three gases is treated with Ammonical AgNO_3 solution to get yellow P.P.T of Ag_2C_2 which are removed. The P.P.T is treated with con. HCl to get C_2H_2 gas.

The left gas mixture is treated with con. H_2SO_4 to get Ethyl hydrogen sulphate ($\text{C}_2\text{H}_5\text{HSO}_4$) which is removed and heated at 16.5°C we get C_2H_4 . The left gas is no doubt C_2H_6 .